BrickPlan 1.0: Certified Assembly-Order Diagnostics for Future Feasibility and Recovery

Release manuscript

September 30, 2026

Abstract

An assembly action can be locally legal and geometrically correct while forcing later removal. End-to-end success alone does not identify this failure mode. BrickPlan 1.0 connects interactive assembly outcomes to exact future-feasibility labels in a tractable, explicitly restricted workcell. The release contains two original LDraw subassemblies, eight episodes, and 26 diagnostic responses. Exhaustive enumeration covers 58,616 reachable states and 568,648 directed transitions. Seven diagnostic dimensions measure dead-end choices, supplied-plan conflict prediction, plan survival, regret, recovery, replanning efficiency, and counterfactual consistency. Every repair start requires at least three removals. A shared HTTP executor supports model clients, human practice, and original-mesh 3D replay. Software controls validate the implementation; no language-model or human performance is reported. This small known-pose release establishes an auditable measurement contract, not broad long-horizon generalization.

1 Motivation and research question

Brick assembly requires choosing an order as well as a final configuration. Closing a region too early can obstruct a later insertion even when the closure is at its correct final pose. A useful planning diagnostic must distinguish this event from an invalid immediate action, a pose-estimation error, or an incomplete response.

Existing studies motivate this distinction but report different quantities. LEGO-Puzzles studies multistep reasoning and planning [1]; PhyBlock evaluates physical planning and understanding [2]; Brick-Composer examines fine-grained part selection and pose prediction [3]. Their planning, selection, and joint step metrics are not interchangeable episode success rates. Interactive disassembly and reconstruction also precede this work, including LTRON [4] and instruction-memory approaches such as InstructioNet [5]. We do not claim that interaction, removal, or long task sequences are new in themselves.

The operational question is: *Does an agent choose actions that preserve completion feasibility, and can it detect and repair a loss at a justified cost?* Version 1.0 deliberately controls placement geometry to isolate order selection. Its contribution is an executable diagnostic contract linking local choices and recovery to certified costs, with explicit missing-response and censoring rules. Establishing external validity and model discrimination remains future experimental work.

2 Workcell and task construction

2.1 Declared action domain

Each task contains a connected, clamped fixed base, a finite inventory, calibrated sites, and a visible target. Every site preserves an original source pose. `attach` and `detach` translate one part along its declared straight path. The workcell includes transfer to an external magazine; motion outside the audited insertion/extraction segment and hand/tool geometry are not modeled. All accepted states remain connected to the fixed base through audited stud connections, treated as rigid holds. The support plate in the renderer denotes this abstract fixture.

The executor uses only current occupancy, path blockers, and connector support. It receives no goal or optimal policy when deciding local legality. It therefore accepts a legal closure even if that choice causes later rework. All public sites remain selectable; there is no next-step action mask. The goal requires exact site occupancy. Equivalent inventory instances share a part type and color, while public instance/site IDs are stable within an episode and permuted by seed.

Each structural attempt costs one action, including failures. Inspection and diagnosis have separate quotas, and finishing costs no structural attempt. The browser supports free camera inspection and exploded target reference. Actual assembly states are never exploded. This interface is for structured practice and review; matched vision-only model/human evaluation is not implemented.

2.2 Geometric qualification

Original LDraw triangles are used for display. Collision qualification uses versioned closed proxy meshes with a 0.25 LDraw-unit inset and 0.002-unit guard. Endpoints are checked for intersection and containment. A clear

Table 1: Frozen structures. Both derive from the same LEGO 31027 provenance group.

Property	Roadster	Kart
Source parts	37	25
Fixed / movable	20 / 17	7 / 18
Reachable states	14,012	44,604
Directed transitions	131,824	436,824
Optimal full actions	17	18
Mandatory chain (parts)	4	4
Minimum repair removals	3	3
Endpoint pairs	666	300
Motion pairs (blocked)	612 (30)	432 (19)
Unknown geometry	0	0

path is certified by separating swept bounds or a distance cover: translation distance is 1-Lipschitz, so a midpoint clearance larger than half the interval length plus the guard covers the whole interval. A triangle-intersection witness certifies a blocked path. Unresolved intervals remain `UNKNOWN` and exclude a case from this release.

One upstream 3022 proxy has a missing triangular stud-cap face. The repair checks its location and boundary against the original source, then fills that face without moving vertices. Source/render geometry is unchanged. These are guarded floating-point certificates over specified meshes and paths, not unrestricted motion or clutch-force guarantees.

2.3 Finite-state certification

Let s denote occupied mutable sites and g the exact target. Exhaustive search enumerates the reachable component. Every legal structural edge has a verified reverse edge. Goal-rooted breadth-first search gives the exact remaining action cost $C^*(s, g)$. Every table is checked against Bellman inequalities, unique zero-cost termination, and a decreasing successor at each nonterminal state. Certificates include geometry, transition, task, and cost hashes.

Let $\Delta = |g| - |s|$. For successful one-part unit-cost actions, attachments minus detachments equal Δ. Consequently the minimum detach count is

$$D^*(s, g) = \frac{C^*(s, g) - \Delta}{2}.$$

No-undo feasibility $V^+(s, g)$ holds exactly when $s \subseteq g$ and $C^*(s, g) = \Delta$. Remaining-budget feasibility is $V_B(s, g, b) = [C^*(s, g) \leq b]$. A no-undo dead end is recoverable with removal; it is not absolute impossibility in the reversible graph.

2.4 Frozen episodes and probes

Each structure supplies full build, checkpoint continuation, injected repair, and matched healthy continuation episodes. Checkpoints follow three legal attachments. Repair starts are reachable and have $D^* = 3$. Healthy controls match installed count and inherit the repair budget, preventing a diagnosis shortcut from the displayed budget. Both are publicly labeled continuation. The default budget is $C^* + \max(4, \lceil C^*/2 \rceil)$.

Two choice probes provide currently legal candidates, with tight remaining budgets equal to the optimal cost. Sixteen lookahead probes use first-conflict distances 1, 2, 4, and 8, with an executable matched-length control per structure and distance. Eight counterfactual responses form one switch pair and one invariance pair per structure. Randomizing IDs and presentation order does not increase the number of independent structures.

3 Seven diagnostic dimensions

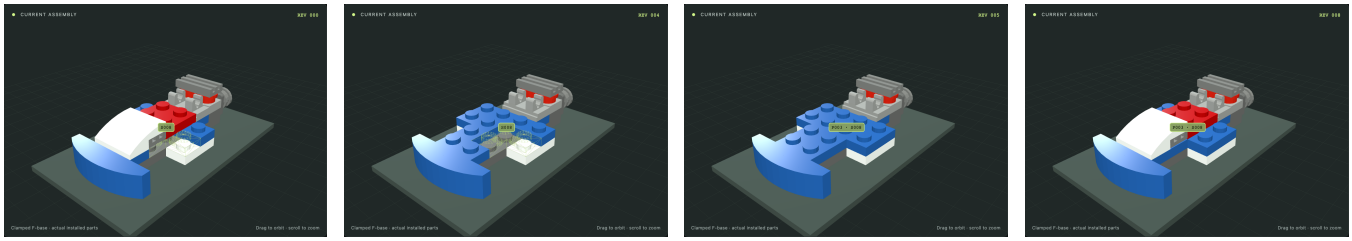
Dead-end choice rate. Among valid assigned legal choices with a feasible predecessor, measure the fraction whose successor loses V^+, and separately the fraction losing $V_B(s', g, b-1)$. Missing/invalid responses are reported separately, along with avoidance over all assigned probes. A model cannot obtain full avoidance credit by abstaining.

Lookahead. Before execution, predict the exact first failing index of a supplied plan or `null` if all its actions execute. Report exact accuracy, conflict recall, false alarms, and coverage by distance. In this release the conflict dependency depth is one; larger distances include intervening legal attachments. Thus these probes measure conflict-prediction horizon, not internal search depth. An executable control need not preserve global completion feasibility.

Plan survival. Record the first attempted-action index losing each feasibility predicate. Recovery does not erase this event. At time t, report known survivors, first-loss outcomes, and unresolved early stops, with bounds $S_{\text{lower}}(t) = N_{\text{alive}}(t)/N$ and $S_{\text{upper}}(t) = (N_{\text{alive}}(t) + N_{\text{unresolved}}(t))/N$. Stratify by structure, mode, reference length, and budget. A completed successful run is absorbing; an abandoned run is not assumed to survive forever.

Regret. For completed successful runs, action regret is attempted actions minus $C^*(s_0, g)$; detach regret is successful detach operations minus $D^*(s_0, g)$. Failed runs receive `null` regret and explicit consumed costs. Under this action domain, action regret equals twice detach regret plus failed attempts. The two cost measures should not be treated as independent latent abilities.

Recovery. Score the first explicit `needsRework` diagnosis before any successful structural change to the injected checkpoint. Earlier failed attempts are allowed and separately flagged. Report detection recall, precision,



(a) Repair start: $D^* = 3$. (b) Remove three closures. (c) Insert P003 at S008. (d) Restore and complete.

Figure 1: Authoritative kart repair replay, seed 31. The replay removes P005, P008, and P018 before inserting the missing interior part. Three removals and four attachments complete the target in seven structural actions, matching the exact optimum. The wireframe in earlier frames denotes the empty calibrated site, not a floating installed brick. These are original source meshes rendered from real executor states; no model performance is implied. Frame revisions are 0, 4, 5, and 8 because diagnosis uses a revision but no structural action.

matched healthy false alarms, completion, and joint detection/completion. Later diagnosis spam cannot replace the first answer.

Replanning efficiency. For successful repair suffixes, report attempted actions, removals, distinct removed instances, excess costs, failed attempts, and the first action reducing D^*. Conditional efficiency must accompany recovery completion rate. Repeated removal counts repeatedly as work but once per distinct part.

Counterfactual consistency. Hold the state, inventory, sites, paths, and two offered closures fixed; change the visible future goal. Both closures are required in every paired target. Certify conditional costs $Q^*(s, a, g) = 1 + C^*(T(s, a), g)$. Switch pairs have different minimizing offered actions; invariance pairs retain the same minimizer. Score joint correctness before raw switching or invariance. The minimum is over offered candidates, not all possible actions: an interior insertion may be globally better. Answers are normalized to physical actions after independent candidate-position permutations.

4 Validation and experimental status

4.1 Evaluation settings

Complete success at the frozen budget is the headline outcome; the seven dimensions form a profile, not a weighted leaderboard score. Policy observations exclude source instruction indices, blockers, future costs, feasibility labels, and witnesses. Private certificate files are for auditors, not model inputs. HTTP requests are revision checked and idempotent. Terminal-only review, exact trajectory replay, atomic journals, and explicit selection among repeated runs make evaluation traceable.

Fourteen adversarial tests cover all frozen goal tables, tampered proofs, missing responses, failed trajectories, censoring, wrong switching, identity equivalence, diagnosis timing, budgets, and replay. A separate actual HTTP test completes eight episodes and 26 probe responses, verifies persistence and restart, rejects duplicate answers, then collects and scores the run. Browser tests complete one full build with a charged incompatible-part attempt and one three-detach repair, then exercise replay, exports, probe identity, and desktop and mobile layouts. Figure 1 uses that certified repair session.

Table 2: Measured software validation controls. These are not VLM baselines.

Controller	Complete	Repair
Privileged optimal oracle	8/8	2/2
Fault + optimal recovery	8/8	2/2
Trap + attach only	2/8	0/2

Table 3: Empirical study slots. No model or human measurements are available.

Study condition	Status
Model, stepwise structured policy	Not evaluated
Same model, explicit planning policy	Not evaluated
Human structured protocol	Not evaluated
Matched image-only protocol	Not implemented

Before a model study, preregister exact model/version, prompt, context and token budgets, reasoning settings, observations, seeds, costs, retry handling, and run-selection rules. Software controls establish internal implementation behavior; they do not demonstrate population-level discrimination or a human advantage.

5 Limitations and next validation target

This release contains two related structures from one source kit. Full optimal lengths of 17 and 18 lie in the proposed D1 range; the longest mandatory chain con-

tains four parts and three edges. It does not establish 64–256-action planning or independence from memorized topology. Lookahead distance is explicitly separated from causal depth. No statistical maximum reasoning horizon is inferred.

Calibrated sites and rigid stud support remove pose estimation and dynamics. Alternative insertion directions, simultaneous multi-part movement, free rotation, grasp geometry, clutch force, gravity-induced failure, and physical construction have not been validated. Certificates apply only to the declared motion domain and proxy geometry. A broader release should vary provenance, dependency depth, true competing plans, and repair severity, while preserving complete certificates where tractable. Real model and human measurements must then test whether these diagnostics explain failures beyond local action errors.

References

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[5] InstructioNet. arXiv:2410.01111, version 1. https://arxiv.org/html/2410.01111v1.